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File Code: 6730 **Date:** August 2, 2016

Subject: Incident Brief/Expanded 72hr report, Pioneer Fire Fuel Geyser Burn Injury

To: Cecilia Seesholtz, Forest Supervisor, Boise National Forest

THE FOLLOWING INFORMATION IS PRELIMINARY AND SUBJECT TO CHANGE

LOCATION: Pioneer Fire, Boise National Forest

DATE OF OCCURRENCE: July 29, 2016

TIME OF OCCURRENCE: Approximately 1400 Hours

ACTIVITY: Fireline Construction, Hazard Tree Removal

NUMBER OF INJURIES/ ILLNESS: One

NUMBER OF FATALITIES: None

PROPERTY DAMAGE: Chainsaw, Nomex Shirt, Nomex Pants, Radio, Line Gear

NARRATIVE:



On Friday, July 29, at approximately 1400, a firefighter received 2nd degree burns to his lower arm when fuel from his chainsaw spewed onto himself and his saw and then combusted.

The firefighter was part of a two-man felling team, working ahead of the crew, falling hazard trees.

When the chainsaw “seemed like it was out of fuel” the sawyer walked away from the fireline and opened the cap to confirm.

That’s when fuel "geysered" from the saw and sprayed the sawyer and his chainsaw. The sawyer’s quick reaction of using his gloved hand to block the geyser protected his face, but directed the fuel along the bottom of his arm and inside the sleeve of his Nomex shirt.

The fuel ignited, engulfing the saw and the sawyer’s arm. The firefighter quickly tossed the saw away when he realized it was on fire. At that moment, he was unaware that he himself was on fire until his swamper pointed it out to him. He then got to a green area, shed his pack, and rolled to extinguish flame.

The Crew EMT responded immediately, providing initial assessment and treatment. It was determined the injured firefighter was able to hike off the fireline, under his own power and meet the Division Paramedic at the road. This hike was estimated to take 20 minutes. As the injured Firefighter and crew EMT were

hiking towards the Paramedic's location they were met by another crew. This crew gave the EMT and the injured firefighter their ATV. The ATV was used to get the injured firefighter to the Paramedic much faster, cutting the time in half. The "9 line" was used to relay information to the team Medical Unit.

Within an hour of the first report, the patient was transported by SUV to a helispot by the Paramedic. Once at the helispot the injured firefighter was flown to helibase by the Pioneer Fire's identified Medivac Helicopter and then loaded onto a commercial air ambulance for transport to the hospital. The firefighter was treated for 2nd degree burns on his lower arm and hand and released from the hospital that same day.

Following the procedures displayed in the 2016 Interagency Standards for Fire and Fire Aviation Operations (The Red Book) Chapter 18, a Lessons Learned Review process has been chosen to fulfill the responsibility to review the accident that occurred on the Boise National Forest on July 29, 2016 by the Forest Supervisor. An Interagency (USFS and BLM) Lessons Learned Review Team has been assembled and was in-briefed on July 31. The specific lessons from this incident will be presented and discussed in the forthcoming Lessons Learned Review report.

In the meantime here are some recent similar events and resources:

Click on the link to view the 2016 WFSTAR: Fireline Fuel Safety

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Homestead Fire Burn Injury, June 16 2016 (2 pages). Burn injuries sustained when fuel erupted from a dolmar and ignited during refueling operations on initial attack.

Fuel Cap Near Miss, May 24, 2016 (4 pages). While no burn injuries occurred on this recent Near Miss, the sawyer's boots and chaps did catch fire. This story provides a good head's up.

Mule Ridge Fire Chainsaw Fuel Geyser, June 6, 2014 (3 pages)

Willow Creek Burn Injury, 2015 (22 pages). A sawyer receives serious burn injuries when fuel sprays out of the chainsaw's gas tank and ignites.

Chainsaw Fuel Mishap - Rock Creek Burn Injury, 2014 (1 page). "A county firefighter received severe burns to the forearms and legs while operating a chainsaw around the perimeter of the fire. The chainsaw was found at the incident site with the fuel cap off and the tank empty. Witnesses saw the fire ignite at ground level and then engulf the firefighter."

Chainsaw Vapor Lock Accident, 2013 (2 pages). A sawyer was injured when gas from a Stihl MS460 chainsaw fuel tank ignited after the sawyer opened the fuel cap to release excess pressure in the tank. The fuel ignited and "flashed" causing minor burns to the sawyer's right arm, right side of neck, and right ear. The sawyer was evaluated by burn center physicians and released. July 1, 2013

Oak Mesa Fire, 2008 (6 pages). "The chainsaw operator thought the saw was running low on fuel due to the way it was running and sounded, and stopped to check. The sawyer stepped back from the line about 5 feet. When the fuel cap was opened, pressurized fuel and vapors vented from the fuel tank, spilling onto the right sleeve of the individual's Nomex fire shirt. He tried to move the chainsaw but it ignited almost immediately and the fire spread instantly to his right arm."

/s/ Jake Strohmeyer

Jake M. Strohmeyer
Learning Review Team Lead

cc: Regional Forester, Randy Draeger, Terry Swinscoe